

IS1105 — Business Information Systems

Full Marking Scheme & Model Answers · Section 1

Module Code	IS1105
Examination Session	Winter 2025
Paper Number	1
Institution	University College Cork
Section Covered	Section 1 (Options 1, 2 and 3 — 50 marks)

Prepared by Study-Uni. This scheme covers Section 1 of the Winter 2025 Paper 1 exam only, as Section 2 (the EDPAC multiple-choice sheet) was not available at the time of preparation. Model answers are provided for all three Section 1 options, since students may select any one.

Question 1(A) — ERP Systems

25 Marks

QUESTION AS SET

Information Systems play a crucial role in organisations, including two key functions: (1) automate transactions and (2) provide information to underpin managerial decision making.

(A) Explain in detail how Enterprise Resource Planning Systems (ERP) provide support for all the transactions in an organisation, discussing the benefits and challenges inherent in their implementation. (25 marks)

FULL MODEL ANSWER

What an ERP system is and why it exists

An **Enterprise Resource Planning (ERP) system** is a single, integrated software platform built around a central shared database that automates and connects the transaction-processing activities of every major function in an organisation — finance and accounting, human resources, procurement, manufacturing, inventory, sales and distribution, and customer service. Before ERP, each department typically ran its own standalone application (a separate payroll system, a separate stock-control system, a separate sales-order system). These "information silos" each captured the same underlying business events — a customer order, a goods receipt, a payroll run — in incompatible formats, so the same transaction had to be re-keyed multiple times and the organisation could never be certain that finance, sales and warehouse all agreed on the true state of the business at a given moment.

How ERP supports transaction processing across the organisation

ERP replaces this fragmented landscape with a single system of record built on a shared, integrated database, and supports transactions in three connected ways:

- **Single point of data entry, many points of use.** A transaction is captured once, at its source (e.g. a sales order keyed in by a sales clerk), and is immediately visible — in real time and without re-keying — to every module that needs it: the warehouse sees the stock has been committed, finance sees the receivable has been created, production planning sees demand against which to schedule a build.
- **Cross-functional process integration.** ERP is organised around end-to-end business processes rather than individual departments — the "order-to-cash" process (order → credit check → pick/pack → ship → invoice → cash receipt) and the "procure-to-pay" process (purchase requisition → purchase order → goods receipt → supplier invoice → payment) each run as a single connected workflow inside the ERP rather than as separate hand-offs between disconnected departmental systems.
- **A common data model and standardised procedures.** Because every module reads and writes to the same underlying tables (the same customer master, the same chart of accounts, the same material master), transaction data is captured in a consistent, standardised format across the whole organisation, which is what ultimately makes company-wide reporting and control possible.

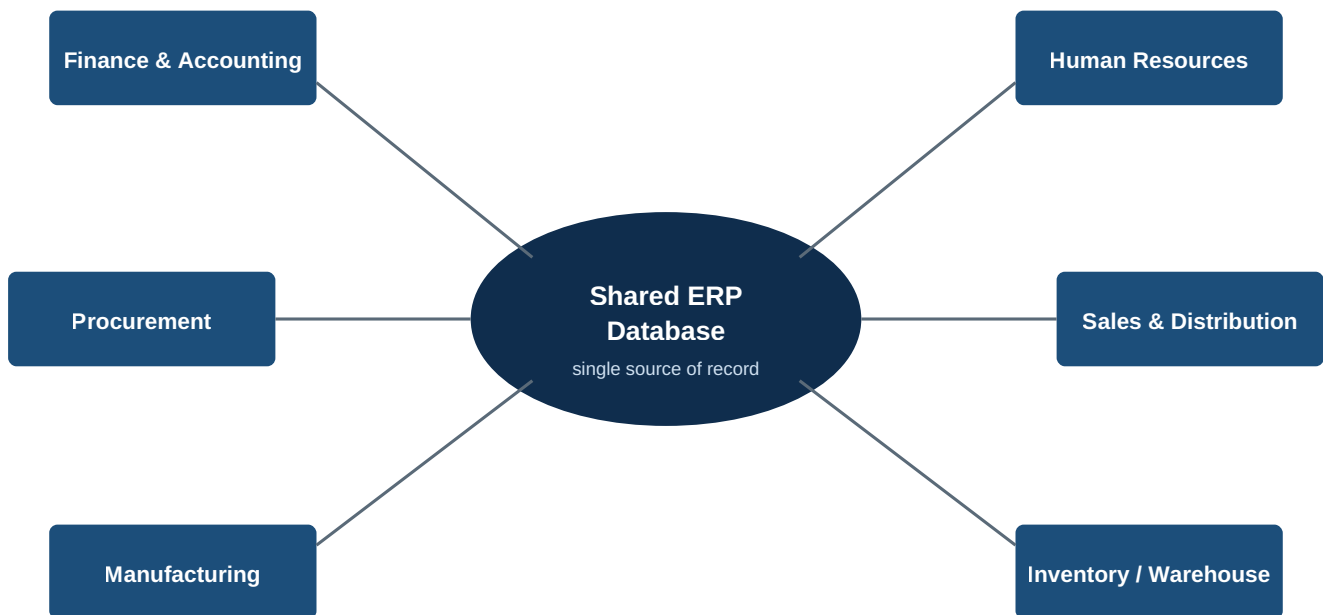


Figure 1 — ERP as a single shared database sitting beneath the organisation's functional modules, replacing siloed departmental systems.

Benefits of ERP for transaction processing

- **Elimination of duplicate data entry and reconciliation effort** — a transaction keyed once is available everywhere it is needed, removing the manual, error-prone reconciliation that siloed systems require.
- **Real-time, organisation-wide visibility** — because every module shares one database, managers can see up-to-date stock levels, order status and financial position at any moment rather than waiting for an end-of-month consolidation.
- **Standardisation and control** — a single set of business rules, approval workflows and a common chart of accounts across every site or subsidiary, which materially eases statutory reporting, audit and regulatory compliance.
- **Operational efficiency** — automating routine transactions (invoice matching, stock replenishment triggers, payroll runs) reduces headcount needed for manual processing and shortens cycle times such as order-to-cash.

Challenges inherent in ERP implementation

- **Cost and duration.** ERP implementations are large, multi-phase projects — covering process redesign, software configuration, data migration and user training — that commonly run from several months to well over a year in a large organisation, with substantial licensing, consulting and internal project-team costs.
- **Business process change and organisational resistance.** Because ERP imposes standardised, cross-functional processes, staff are frequently required to abandon long-standing local ways of working. This change is consistently identified as one of the leading causes of implementation difficulty, and requires sustained senior-management sponsorship and a dedicated, cross-functional project team to overcome.
- **Data migration and integration complexity.** Legacy data is rarely clean, complete or consistently structured, so migrating and validating it into the new common data model is a major source of risk and delay.
- **Customisation versus standardisation trade-off.** Every organisation has processes that do not map neatly onto the ERP vendor's standard configuration; heavy customisation restores flexibility but increases cost, extends

timelines and complicates future upgrades, while accepting the vendor's standard process may require the organisation to change working practices it would rather keep.

- **Risk of implementation failure.** A substantial share of ERP projects fail to meet their original timeline, budget or benefits targets, which can disrupt order fulfilment, cash flow and regulatory reporting during and after go-live — underlining why implementation planning and change management are as important to the answer as the technical benefits.

SOURCING NOTE Named benefit/challenge figures and implementation-risk statistics (e.g. typical implementation duration, failure-rate ranges) are drawn from current ERP-vendor and implementation literature (NetSuite, Oracle) rather than from IS1105 course material, since the course pack does not contain a dedicated ERP lecture. They are standard, widely corroborated figures in the ERP literature, not exam-specific content.

Mark Allocation — Question 1(A)

Band	Descriptor	Marks
Excellent 21–25	Clearly explains ERP as an integrated, single-database platform automating transactions across functions; explicitly links this to cross-functional processes (e.g. order-to-cash, procure-to-pay); gives well-developed, correctly balanced benefits and challenges with justification.	21–25
Good 16–20	Explains ERP's role in automating transactions with reasonable detail; identifies most major benefits and challenges but development or linkage between them is less complete.	16–20
Partial 10–15	Basic, largely descriptive account of ERP; benefits and/or challenges are listed rather than explained; limited evidence of understanding how ERP actually integrates transactions.	10–15
Minimal 0–9	Little or no accurate explanation of ERP; answer is generic, largely irrelevant, or confuses ERP with an unrelated system type.	0–9

Question 1(A) Total: 25 Marks

Question 1(B) — Decision Support Systems

25 Marks

QUESTION AS SET

(B) Explain in detail what is involved in the concept of Decision Support Systems (DSS) and how it can help managers make evidence-based decisions. Illustrate your answer with specific examples of DSS applications in functional areas of your choosing. (25 marks)

FULL MODEL ANSWER

The concept of a Decision Support System

A **Decision Support System (DSS)** is an interactive information system designed to help managers — typically at middle and senior levels — make semi-structured or unstructured decisions that cannot be fully automated, by gathering, analysing and synthesising data into the specific comparisons, projections or scenarios a decision-maker needs. This distinguishes a DSS from a Transaction Processing System (TPS), whose job is simply to capture and store operational data, and from routine Management Information System (MIS) reporting, which produces fixed, scheduled reports: a DSS is built to answer the manager's specific "what if" question at the point a decision actually has to be made.

A DSS is generally understood to have three core components:

- **A database (or data management) component** — pulling together relevant internal transaction data (often sourced from the organisation's ERP or transaction systems) and, where relevant, external data such as market prices or competitor information.
- **A model base** — the analytical engine: statistical, financial, forecasting or optimisation models (e.g. a demand-forecasting model, a what-if budget model, a linear-programming scheduling model) that convert raw data into decision-relevant insight.
- **A user interface** — dashboards, query tools or "what-if" controls that let a non-technical manager interrogate the model base without needing to write code or queries themselves.

How a DSS supports evidence-based decision making

Rather than a manager relying on intuition or partial information, a DSS lets them test alternative courses of action against the organisation's actual data before committing to one. Concretely, a DSS supports evidence-based decisions by allowing the manager to:

- **Run "what-if" scenarios** — e.g. modelling the effect of a price change on projected revenue and margin before the price is actually changed;
- **Compare alternatives side by side** against consistent, shared data and criteria rather than each manager working from a different version of the truth;
- **Forecast and identify trends** that are not visible from a single period's raw transaction data, by applying statistical or predictive models to historical data; and
- **Quantify risk**, for example by producing a range of likely outcomes rather than a single point estimate, so the decision explicitly accounts for uncertainty.

In every case, the underlying logic is the same: the decision is grounded in the organisation's own data and an explicit model, and is therefore auditable and repeatable, rather than resting solely on managerial judgement.

Illustrative examples of DSS applications by functional area

Functional area	Illustrative DSS application	How it supports the decision
Finance	Budgeting and cash-flow forecasting tools; credit-risk and portfolio-analysis models used by banks and treasury functions	Lets a finance manager model different budget or investment scenarios and quantify risk (e.g. probability of a covenant breach) before committing capital, rather than deciding on a single static forecast.
Marketing	Customer segmentation and campaign-planning dashboards built on customer purchase and response data	Allows a marketing manager to target the segment most likely to respond to a specific campaign, using evidence from past purchasing behaviour rather than a generic, un-targeted campaign.
Operations / Supply Chain	Demand-forecasting and inventory-optimisation systems that recommend reorder points and quantities	Supports a decision on how much stock to hold and when to reorder by balancing forecast demand against holding cost and stock-out risk, rather than relying on a fixed reorder rule.
Human Resources	Workforce-planning tools that model staffing levels against projected demand or attrition	Gives HR managers an evidence base for recruitment and scheduling decisions instead of ad-hoc headcount requests from individual departments.

SOURCING NOTE The three-component DSS architecture (database / model base / user interface) and the functional-area examples above reflect standard, widely taught MIS/DSS theory (e.g. as presented by CFI, TechTarget and comparable MIS references) rather than a specific IS1105 lecture slide, since the provided course pack did not include a dedicated DSS lecture. A student citing any two or three well-explained functional examples, rather than all four, should be credited in full provided the explanation of how the example supports evidence-based decision-making is genuinely developed.

Mark Allocation — Question 1(B)

Band	Descriptor	Marks
Excellent 21–25	Clearly explains the DSS concept (components and purpose) and explicitly connects it to evidence-based decision making; provides at least two well-developed, correctly explained functional-area examples showing genuine understanding of how the DSS informs the decision.	21–25
Good 16–20	Reasonable explanation of DSS and its link to decision-making; at least one functional example given, but development of either the concept or the example(s) is less complete.	16–20
Partial 10–15	Basic or partial definition of DSS; weak or missing link to evidence-based decision-making; examples, if present, are generic or unexplained.	10–15
Minimal 0–9	Little or no accurate explanation of DSS; no relevant functional example provided.	0–9

Question 1(B) Total: 25 Marks | Question 1 Total: 50 Marks

Question 2 — Database Types and Characteristics

50 Marks

QUESTION AS SET

Databases are important resources for managing an organisation's data resources. Critically discuss the key characteristics of different types of databases for processing transactions, including how data is stored and processed. (50 marks)

FULL MODEL ANSWER

Why databases matter for transaction processing

A database is an organised, centrally managed collection of data that removes the redundancy and inconsistency inherent in older file-based systems, where each application kept its own separate data files. Because a database is accessed through a Database Management System (DBMS) rather than directly by individual applications, the organisation gets three things a file-based approach cannot easily provide: reduced data redundancy (the same fact is stored once, not copied across many files), improved data integrity and consistency (validation rules are enforced centrally), and controlled, shared access by many users and applications at once — which is precisely what high-volume transaction processing requires.

Relational databases

The dominant model for transaction processing is the **relational database**. Data is stored in tables (relations) of rows and columns, where each table represents one entity (customers, orders, products) and relationships between tables are expressed through shared key fields (a foreign key in the Orders table referencing the primary key of the Customers table) rather than by physically nesting or duplicating data. Data is retrieved and manipulated using Structured Query Language (SQL). Relational databases enforce strict rules (referential integrity, normalisation to remove redundant duplication) that make them well suited to **OLTP — Online Transaction Processing**: high volumes of short, well-defined transactions (place an order, update a stock balance, post a payment) that must complete reliably and consistently. This reliability is formalised through the **ACID properties**: Atomicity (a transaction either completes entirely or not at all), Consistency (the database moves from one valid state to another), Isolation (concurrent transactions do not corrupt each other), and Durability (once committed, a transaction survives a subsequent system failure). Examples in wide organisational use include Oracle Database, Microsoft SQL Server, MySQL and PostgreSQL.

NoSQL / non-relational databases

Where relational databases assume rigid, pre-defined table structures, **NoSQL databases** are designed for large volumes of unstructured or semi-structured data and for horizontal scalability across many low-cost servers rather than strict transactional consistency. Several sub-types exist:

- **Document databases** (e.g. MongoDB) store data as flexible, self-contained documents (typically JSON-like), well suited to data whose structure varies between records, such as product catalogues.
- **Key-value stores** (e.g. Redis) store data as simple key/value pairs, optimised for extremely fast lookups, such as session data or a shopping-cart cache.

- **Column-family stores** (e.g. Apache Cassandra) organise data by column rather than row, suited to very large-scale, write-heavy workloads such as sensor or log data.
- **Graph databases** (e.g. Neo4j) store data as nodes and the relationships between them, suited to highly interconnected data such as social networks or fraud-detection link analysis.

NoSQL databases typically favour "eventual consistency" over strict ACID guarantees in exchange for scalability and flexibility, so they are usually chosen for high-volume, variably structured data (web logs, social media content, IoT sensor streams) rather than for core financial transactions where strict consistency is essential.

Data warehouses (OLAP)

A **data warehouse** is a separate, large-scale repository that consolidates historical data drawn from multiple operational (OLTP) systems, structured specifically for analysis rather than for processing new transactions. Its usage pattern is **OLAP — Online Analytical Processing**: complex, exploratory queries across large volumes of historical data (e.g. "compare quarterly sales by region and product category over the last five years") rather than the short, high-frequency read/write transactions a relational OLTP database handles. Data is commonly organised in a **star schema** — a central fact table of measurable events (e.g. individual sales) surrounded by dimension tables (time, product, customer, region) that describe the context of each fact — which is optimised for fast aggregation and slicing rather than for the transactional integrity a relational OLTP system prioritises.

Comparative summary

Database type	How data is stored	How data is processed	Best suited to
Relational (OLTP)	Structured tables of rows/columns linked by keys	SQL queries; ACID-compliant transactions	High-volume, well-defined operational transactions (orders, payments, stock movements)
NoSQL	Documents, key-value pairs, column families, or graphs — schema-flexible	Distributed, horizontally-scaled reads/writes; typically eventual consistency	Large volumes of unstructured/semi-structured or fast-changing data (web, social, IoT)
Data warehouse (OLAP)	Consolidated historical data, often in a star schema of facts and dimensions	Complex multidimensional aggregation and trend queries	Management reporting, business intelligence, trend and pattern analysis

Critical discussion

No single database type is universally "best" — the appropriate choice depends on the nature of the workload. Relational databases remain the correct default for core transaction processing because their strict consistency guarantees are exactly what financial and operational transactions require: a bank cannot tolerate an "eventually consistent" account balance. Their limitation is that rigid schemas and vertical scaling make them comparatively costly to scale to the volumes and variety of unstructured data generated by modern web and IoT sources, which is precisely the gap NoSQL databases were designed to fill — at the cost of the strict integrity guarantees a relational system provides. A data warehouse addresses a different problem again: relational OLTP databases are deliberately optimised for fast, narrow transactions, which makes them a poor fit for the broad, exploratory historical queries decision-makers need, so organisations typically operate a relational database for day-to-day transaction processing *and* a separate data

warehouse, periodically populated from it, for analysis and decision support — directly connecting this question to the role of DSS discussed in Question 1(B).

CROSS-REFERENCE See Question 1(B): a data warehouse is frequently the database component that underpins a DSS's model base, since DSS analysis draws on consolidated historical data rather than live transactional records.

Mark Allocation — Question 2

Band	Descriptor	Marks
Excellent 41–50	Accurately explains at least two distinct database types (must include relational) with correct detail on storage structure and processing method (e.g. ACID, SQL, schema flexibility); genuinely critical comparison identifying strengths, limitations and appropriate use-cases; well organised with clear structure (e.g. comparative table).	41–50
Good 31–40	Correctly explains relational databases and at least one alternative type with reasonable detail on storage/processing; some comparative or critical discussion present but less fully developed.	31–40
Partial 18–30	Covers relational databases only, or covers multiple types at a superficial/descriptive level; little genuine critical comparison; storage vs. processing distinction unclear.	18–30
Minimal 0–17	Little or no accurate explanation of database types; answer is generic or largely irrelevant to transaction processing.	0–17

Question 2 Total: 50 Marks

Question 3 — The Role of AI in Society

50 Marks

QUESTION AS SET

Using examples, discuss the role of artificial intelligence (AI) in society. (50 marks)

FULL MODEL ANSWER

What AI is, in brief

Artificial Intelligence (AI) refers to computer systems capable of performing tasks that would normally require human intelligence — recognising patterns, making predictions, understanding language, or making decisions — typically by being trained on large volumes of data rather than being explicitly programmed with fixed rules for every situation. Machine learning (systems that improve their performance from data) and, more recently, generative AI (systems that produce new text, images or code) are the two forms most visible in society today. An excellent answer should discuss AI's role across several distinct areas of society, using specific, well-explained examples in each, and should weigh benefits against genuine societal challenges rather than treating AI as unambiguously positive.

Business and the economy

Within organisations, AI is embedded in decision support and automation: recommendation engines (e.g. those used by Netflix and Amazon to personalise content and product suggestions from a user's viewing or purchase history), fraud-detection systems in banking that flag anomalous transactions in real time, demand-forecasting tools that extend the DSS concept discussed in Question 1(B) by using machine-learning models rather than purely statistical ones, and generative AI tools increasingly used for drafting, coding assistance and customer-service chatbots. This extends automation beyond the structured, rule-based transaction processing of an ERP system (Question 1(A)) into tasks that previously required human judgement.

Healthcare

AI is used to support (not replace) clinical decision-making: image-analysis systems assist radiologists in detecting patterns in scans, and predictive models help hospitals identify patients at elevated risk of deterioration or readmission, supporting earlier intervention. These systems function as a form of decision support for clinicians, analogous to the DSS concept, but applied to diagnostic and treatment data.

Transport and everyday life

Navigation and mapping services (e.g. Google Maps) use AI to predict traffic and route journeys in real time; increasingly autonomous and driver-assistance features in vehicles use AI for object recognition and hazard detection; and voice assistants and smart-home devices use AI for natural-language understanding, illustrating how AI has moved from specialist organisational use into everyday consumer life.

Public services and governance

Public bodies increasingly use AI-assisted systems for tasks such as detecting fraudulent welfare or tax claims, triaging citizen service requests, and forecasting demand for services such as emergency care. This raises AI's role from a

purely commercial tool to one with direct implications for how public resources are allocated and how citizens are treated by the state.

Societal challenges and risks

A genuinely critical answer must weigh these benefits against real risks:

- **Bias and fairness.** Because AI systems learn from historical data, they can reproduce and amplify existing societal biases (for example, biased outcomes in AI-assisted recruitment or credit-scoring systems that disadvantage particular groups), which is a well-documented and actively researched problem rather than a hypothetical one.
- **Employment and job displacement.** Automation of both routine and increasingly cognitive tasks raises legitimate concern about job displacement in affected sectors, alongside the creation of new roles, meaning the net societal effect is genuinely contested rather than settled.
- **Privacy.** AI systems typically depend on large volumes of personal data, raising concerns about surveillance, data protection and consent that regulators are actively responding to.
- **Accountability and transparency.** Many AI models (particularly deep-learning systems) are difficult to fully explain, which is problematic when they inform consequential decisions such as loan approval, medical diagnosis or criminal-justice risk assessment, and is a central motivation behind emerging regulation.
- **Regulatory response.** Governments have begun legislating specifically in response to these risks — most notably the EU's Artificial Intelligence Act, which classifies AI applications by risk level and imposes the strictest obligations on "high-risk" uses such as those in employment, credit and law enforcement — illustrating that AI's societal role is now actively being shaped by policy, not left purely to the market.

Conclusion

AI's role in society is now pervasive rather than niche: it sits inside business decision-making, healthcare, transport, public services and everyday consumer technology simultaneously. Its societal value depends on treating it as a tool that augments and supports human decision-making — consistent with the decision-support role discussed elsewhere in this paper — rather than as an unaccountable replacement for human judgement, and its continued adoption is now being actively shaped by regulation designed to manage its risks alongside its benefits.

SOURCING NOTE Named real-world examples (Netflix/Amazon recommendation systems, Google Maps routing) and the reference to the EU AI Act are drawn from current, independently verifiable public reporting rather than from a specific IS1105 lecture, since AI is treated at survey level (Lecture 4) in the course pack. A student may substitute equally well-explained alternative real examples in each category and should be credited equivalently.

Mark Allocation — Question 3

Band	Descriptor	Marks
Excellent <i>41–50</i>	Discusses AI's role across multiple distinct areas of society (e.g. business, healthcare, transport, public services) with specific, well-explained, correct examples in each; explicitly and critically discusses genuine societal challenges (bias, privacy, employment, accountability/regulation) alongside benefits; coherent, well-organised argument.	41–50
Good <i>31–40</i>	Covers at least two areas of society with relevant examples; some discussion of societal challenges present but less developed or one-sided.	31–40
Partial <i>18–30</i>	Covers AI examples in a single area only, or examples given without real explanation; little or no discussion of societal challenges/risks.	18–30
Minimal <i>0–17</i>	Little or no accurate discussion of AI; no relevant examples; answer largely generic or off-topic.	0–17

Question 3 Total: 50 Marks